LangChain

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LangSmith Self-hosted

Deployment, Adoption, and Scale.

Tom Straub • Architect • Tuesday, October 27th, 2026

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Today

Agenda

- LangSmith Platform Architecture
- Terraform Module Review Agent Development
- Friction Log
- Q & A

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Platform Architecture

Self-hosted LangSmith on a dedicated GKE cluster.

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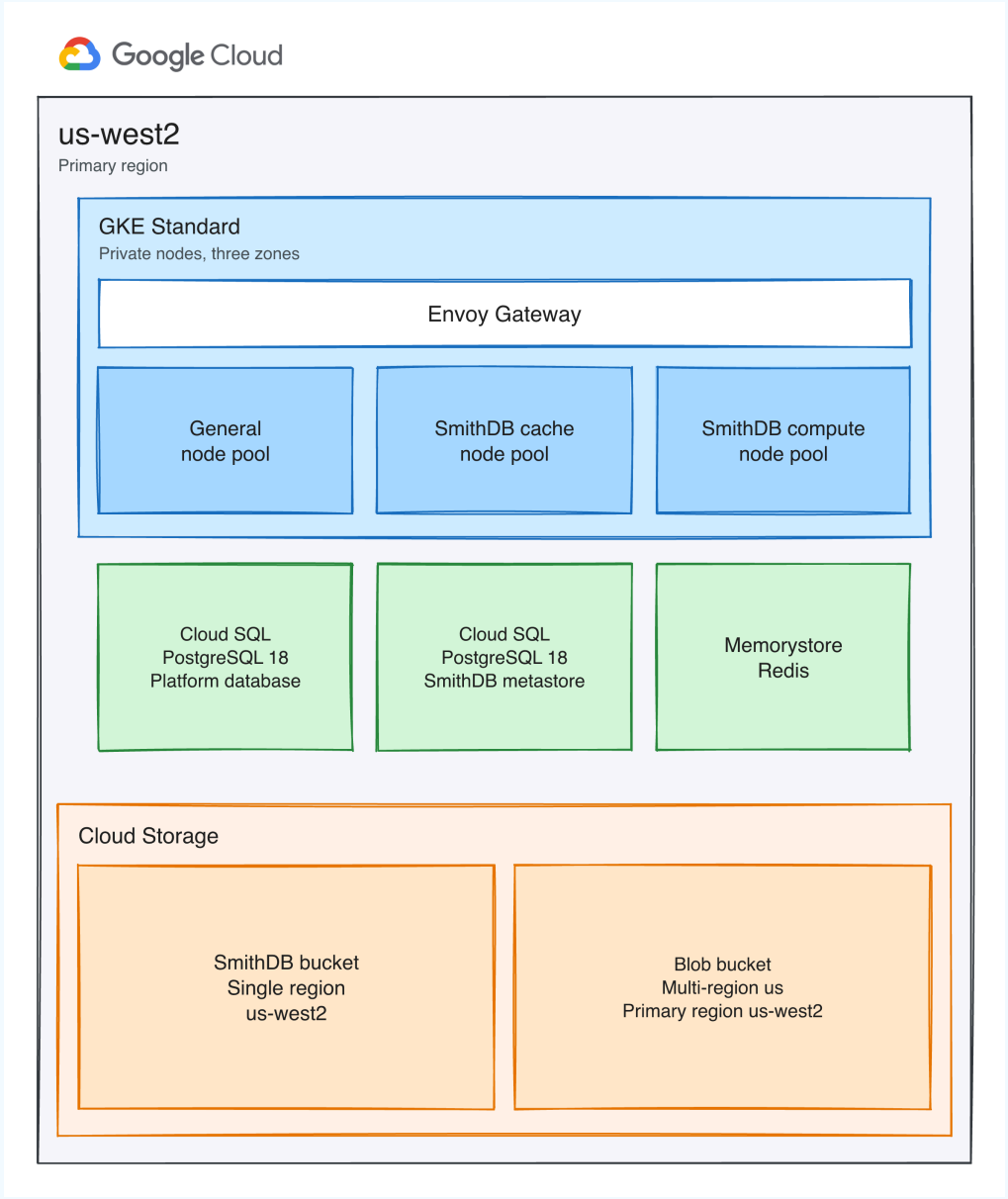
Platform

What we designed around

- A dedicated GKE Standard cluster.
- Agents stay where they are and send traces to LangSmith.
- Durable data sits in managed services, outside the cluster.
- A standard install now. Hosted agents come later.

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Architecture



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Platform

Infrastructure

Application

A dedicated GKE Standard cluster.

- General node pool
- SmithDB Cache node pool
- SmithDB Compute node pool

Storage

- Cloud SQL - Users, projects, datasets, experiments, and Chat.
- Blob GCS bucket - Payloads and attachments.
- Redis - Ingest queue, cache, and Chat.
- SmithDB Cloud SQL - Metastore, empty and separate.
- SmithDB GCS bucket - Traces and feedback.

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Platform

Scaling

The size we start at

About 20 concurrent users and 100 traces a second, with bursts toward 1,000

The size we move to

About 50 concurrent users and 1,000 traces a second, when the first ranges stay full.

- Redis - Grows with users.
- Blob GCS bucket - Grows with payloads.
- SmithDB Cloud SQL - A larger metastore.
- SmithDB Cache and compute - Larger machines.
- SmithDB GCS bucket - Grows with retained traces.

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BYOC

If operating LangSmith yourself is more than you want, we can talk about "Bring Your Own Cloud"

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Agent Development: Terraform Module Review

A module review that stops before it opens a pull request.

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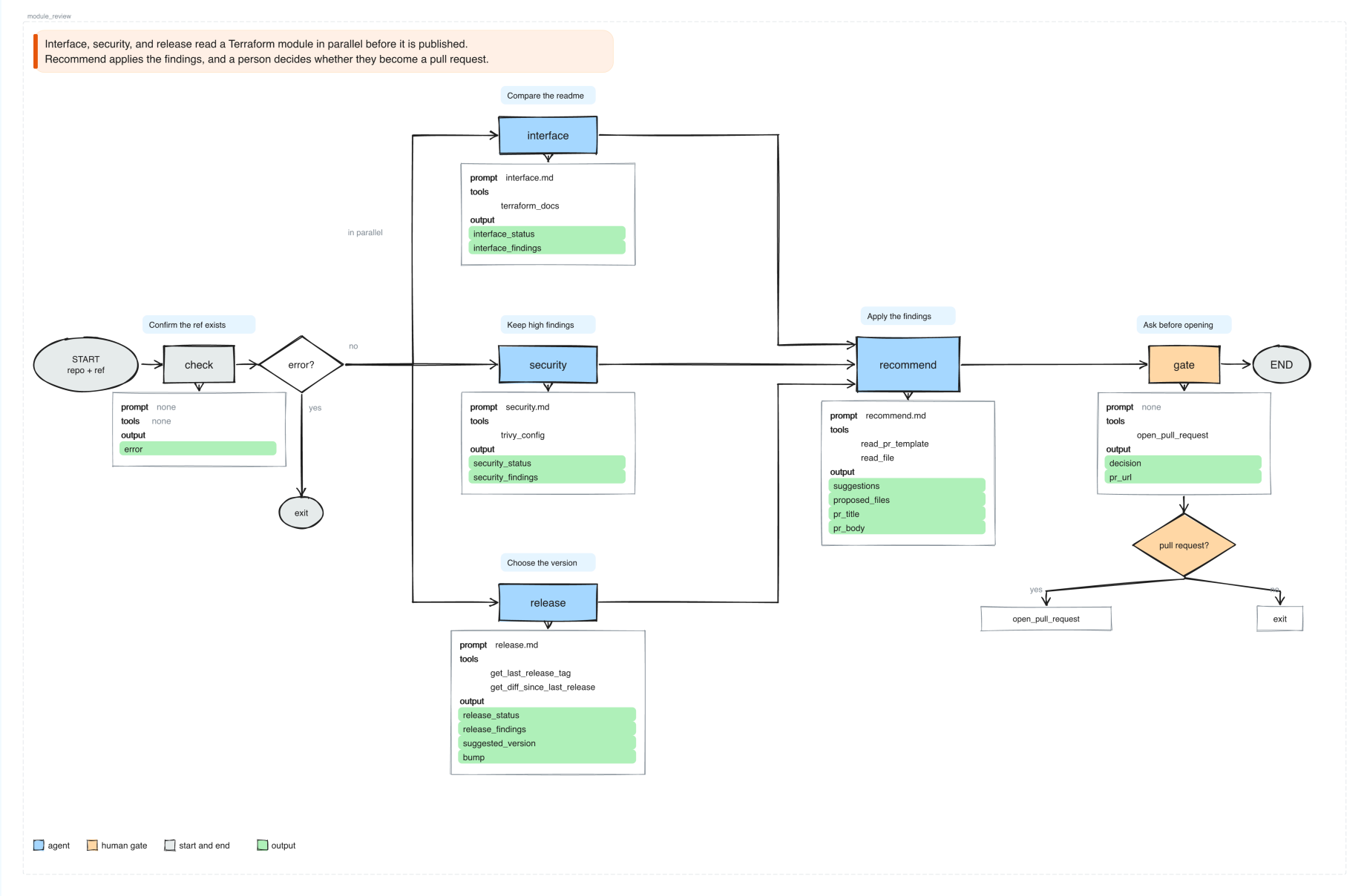
Agents

What Was Built

- The work is a Terraform module that is ready to publish.
- Interface, security, and release read the repo in parallel.
- Recommend drafts the files and the pull request.
- A person answers yes or no before a Pull Request is created.

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Agent Architecture



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Agents

Evaluation Results

The set

Seven frozen modules.

The agents call the same tools and receive canned files and diffs.

The bar

Five scores, each pass or fail.

A scenario counts only when all five pass.

The cost

Median time and tokens for each agent, read off the latest experiment.

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Agents

Business Value

Self-service

The module is the interface teams use to build infrastructure.

Clear variables and usable settings are what let that scale.

Interface gaps become edits on the next release.

The right bump

Major, minor, and patch are applied inconsistently

Every candidate gets the same recommendation from the org rule, before the tag.

Caught in development

The same checks run on every upcoming release.

A person approves the pull request. Consumers do not find the miss.

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Friction

Lessons learned.

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Friction

What got in the way

- The eval stops at the human gate, so the pull request is never scored.
- One drifted row fails the suite. The previous experiment is another view.

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Friction

What we would ask for

- Let a dataset example carry a scripted yes or no for the interrupt.
- A scorer that checks the required tool calls, not only the final fields.
- A score for the rewritten file against the finding it was meant to fix.

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Questions?

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Thank you

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